Design Brief – Review 2

0. Header

Project code	FA26SE238
Project name	DroneOps Intelligence
Team	GFA26SE167
Supervisor	Pham Minh Tri; Dang Ngoc Minh Duc
Repository link (tag review-2)	https://github.com/alexLe2802/DroneOps-Intelligence- (commit 48aea3c; review-2 tag not present)
Version	Review 2 submission v1.0
Date	27 October 2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
AI decision support	Top-k historical similarity retrieval with evidence	Configured external AI advisory service; remove mandatory vector retrieval	Iteration 1 technology/data constraint; requires Review 2 approval
Mission changes	Mission versions	One current mission plus separately reviewed update requests	Iteration 1 finding: prevents rejected edits overwriting approved content
Client platforms	Web and Mobile App	Web platform for Review 2 baseline	Team capacity and no mobile implementation baseline; requires Review 2 approval

1.2 Out of Scope

Item	Reason
Low-level flight control/autopilot	The product manages mission operations and never commands vehicle stabilization/navigation.
Aviation certification and unprovided restrictions	Validation uses configured data and is not a legal airspace guarantee.
Custom AI model training	The project integrates a configured provider and retains human decision authority.

2. System at a Glance

DroneOps Intelligence is a Web mission-operations platform for UAV operators and operations managers. It centralizes aircraft, operator, mission and restriction data; validates planned routes before approval; ingests MAVLink-derived telemetry; detects and manages incidents; and preserves post-flight evidence. Operators plan and monitor missions, while managers review proposals, restrictions and operational outcomes. Advisory AI summarizes authorized context and suggests risks and precautions without approving missions or controlling a UAV. The core value is a traceable workflow from planning through approval, monitoring and post-flight review, replacing fragmented maps, logs and spreadsheets. System structure and trust boundaries are shown in Report 4, SDD §1.1, Figures 1–2.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	*Use layered Web/API/domain/persistence architecture*	*Monolith; microservices; layered modular application*	*Layered modular application matches four-person delivery while keeping business and persistence boundaries testable.*	*Le Quoc Thong*	*All FRs; SDD Figures 1–4*
D2	*Review mission edits before applying them*	*In-place edit; immutable versions; pending update request*	*Pending requests preserve approved content and allow atomic accept/reject on the same mission ID.*	*Ha Quang Nghia*	*FR-05..10; SDD Figures 6–12*
D3	*Normalize telemetry at a gateway boundary*	*Direct browser link; broker-only; authenticated gateway/API*	*The gateway isolates MAVLink transport and supplies idempotent normalized samples to the API.*	*Le Quoc Thong*	*FR-11..13; SDD Figures 14–16*
D4	*Persist post-flight evidence and coverage*	*Ephemeral dashboard; raw logs only; reports over stored evidence*	*Stored samples/incidents support scoped playback, reproducible metrics and explicit missing coverage.*	*Dang Dao Anh Duc*	*FR-17..18/22; SDD Figures 17–18*
D5	*Keep AI advisory and fail-open for manual review*	*Autonomous approval; local model; configured external API*	*External API reduces model scope; provenance and failure states preserve human authority and mission availability.*	*Giang Nguyen Gia Khanh*	*FR-19; SDD Figure 19*

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	*Registered Top-k retrieval is not implemented.*	*Defense scope may be incomplete.*	*Obtain BM03 decision; if retained, implement labeled k=5 retrieval and Precision@5 evidence.*	*Giang Nguyen Gia Khanh*
R2	*Realtime/staging workflow evidence is incomplete.*	*Monitoring or incidents may fail during defense.*	*Run deterministic gateway replay, disconnect/reconnect, acknowledgement and resolution scenarios.*	*Le Quoc Thong*
R3	*Design/schema/state documentation differs from code.*	*Reviewers cannot trace behavior reliably.*	*Regenerate ERD/state mappings from migrations and canonicalize status values.*	*Ha Quang Nghia*

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G-01	Registered Top-k similarity/vector retrieval is specified in the SRS but has no implementation in the source; the implemented OpenAI adapter is not a substitute for retrieved historical evidence.	Registration §3.2(b–g); SRS FR-19/NFR-M02; source	Planned – Giang Nguyen Gia Khanh – 30 Nov 2026; implement and evaluate retrieval unless BM03 change is accepted.	Giang Nguyen Gia Khanh	30 Nov 2026
G-02	The project material identifies a Mobile App/operator mobile client, but the repository contains only the Next.js Web	Report 1 §1.1/§7; SRS mobile statements; source	Planned – Le Quoc Thong – 30 Nov 2026; remove mobile scope only if BM03 change is	Le Quoc Thong	30 Nov 2026

ID	Gap	Location	Plan	Owner	Due
	client and no defined mobile feature subset or mobile package.		accepted, otherwise define and deliver its subset.		
G-03	The logical ERD/entity catalogue omits implemented records including MissionUpdateRequest, MissionValidationRun and DroneConnection, so design-to-schema traceability is incomplete.	SDD §2 ERD/entity table; Domain/AppDbContext	Planned – Ha Quang Nghia – 06 Nov 2026; update ERD and entity catalogue from the verified schema.	Ha Quang Nghia	06 Nov 2026
G-04	The four main workflows have designed cases but no recorded staging end-to-end execution evidence or pass results.	SRS main workflows; Report 5; CI	Planned – all workflow owners – 30 Nov 2026; automate and retain seeded staging evidence.	All workflow owners	30 Nov 2026
G-05	Mission status spelling is not canonical across code and documents (for example InProgress and In Progress), which can create undocumented state variants.	SRS BR-13; state diagrams; application services	Planned – Ha Quang Nghia – 06 Nov 2026; introduce canonical state constants and align diagrams/migrations.	Ha Quang Nghia	06 Nov 2026

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	*Identity, operator and UAV management*	*Operator; Manager*	*An authorized user can access only permitted profiles/UAVs; invalid or out-of-scope create/update/read requests are rejected and audited.*	*TC-F01-01..03*	*In progress*
F-02	*Mission planning, validation and approval*	*Operator; Manager*	*A route is checked at waypoint and segment level; only a current valid request can be approved; rejection leaves current mission content unchanged.*	*TC-F02-01..03*	*In progress*
F-03	*MAVLink telemetry and live monitoring*	*Operator; Manager; Gateway*	*Authenticated normalized samples are persisted idempotently and visible with source time, freshness, location, altitude, speed, battery and progress within the NFR threshold.*	*TC-F03-01..03*	*In progress*
F-04	*Operational incident lifecycle*	*Operator; Manager*	*Configured abnormal conditions create one traceable incident episode; acknowledgement and resolution retain actor, time, reason and frozen evidence.*	*TC-F04-01..03*	*In progress*
F-05	*Post-flight history, reports and analytics*	*Operator; Manager*	*A completed mission exposes scoped history, route playback, incidents, coverage and calculated metrics; missing evidence is shown as unavailable.*	*TC-F05-01..03*	*In progress*
F-06	*AI-assisted mission assessment*	*Operator; Manager*	*Authorized mission context returns traceable advisory output or an explicit unavailable state; provider failure never blocks manual review and AI never approves a mission.*	*TC-F06-01..03*	*In progress*

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	API performance	95th percentile list/detail latency	<= 2 s under P-01	Run >=100 timed requests after warm-up; record hardware/network.
NFR-02	Realtime performance	Gateway receipt-to-live-view latency	95% <= 2 s	Correlate synchronized gateway-received and client-rendered timestamps.
NFR-03	Validation performance	Pre-flight validation duration	95% <= 3 s	Run >=30 controlled valid/invalid routes under P-01.
NFR-04	Reliability	Monitoring continuity and persisted-record loss	60 min; 0 crash; 0 acknowledged record loss	Controlled soak test plus database and trace comparison.
NFR-05	Security	Unauthorized/out-of-scope operations rejected	100% of authorization matrix	Positive/negative role and direct-ID API tests.
NFR-06	Usability	Representative users completing prepared mission task	>=4/5 within 5 min after <=30 min onboarding	Timed task study with a prepared five-waypoint mission.

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Le Quoc Thong (SE172002)	Leader; frontend/dossier; MF2	Web integration; gateway/live monitoring; CI/documentation	WF-02	D1, D3; current implementation evidence to verify
Ha Quang Nghia (SE184490)	Backend/MF1 owner	Mission review; validation; schema/state alignment	WF-01	D2; mission workflow services
Giang Nguyen Gia Khanh (SE182367)	AI/MF4 owner	AI context; provider adapter; advisory/failure handling	WF-04	D5; AI assessment evidence to verify
Dang Dao Anh Duc (SE194009)	Post-flight/MF3 owner	History; reports; analytics/playback	WF-03	D4; post-flight services

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Mission Planning and Pre-flight Validation	Operator; Manager	Operator submits a new or changed route.	A reviewed, safe plan is approved or rejected with evidence.	F-01, F-02	Mission: Draft -> Pending Review -> Approved/Rejected	Ha Quang Nghia
WF-02	Real-Time Monitoring and Incident Management	Gateway; Operator; Manager	An approved mission starts and telemetry arrives.	Live state is visible and abnormal events are handled traceably.	F-03, F-04	Mission: Approved -> In Progress; Incident: Open -> Acknowledged -> Resolved	Le Quoc Thong
WF-03	Post-Flight Analysis and Mission History	Gateway; Operator; Manager	A trusted completion event is recorded.	Users review route, evidence, incidents, report and analytics.	F-05	Mission: In Progress -> Completed; Report: Queued -> Generating -> Available/Failed	Dang Dao Anh Duc
WF-04	AI-Assisted Mission Risk Assessment	Operator; Manager; AI Provider	An authorized user requests an assessment.	Advisory risks and precautions or an explicit failure state support human review.	F-06	Assessment: Pending -> Available/Failed; Mission unchanged	Giang Nguyen Gia Khanh

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	*WF-01*	*Main*	*Seeded Operator, Manager, UAV and valid restriction set*	*Create route; validate; submit; Manager reviews and approves*	*Same mission becomes Approved with decision and current evidence*	*Ha Quang Nghia*
DS-02	*WF-01*	*Exception*	*Approved mission and invalid proposed route*	*Submit edit; validation fails; attempt approval/reject*	*Approval blocked or proposal rejected; approved content is unchanged*	*Ha Quang Nghia*
DS-03	*WF-02*	*Main*	*Approved active mission; authenticated gateway replay*	*Stream normal/abnormal samples; acknowledge and resolve incident*	*Live view updates and complete incident timeline is retained*	*Le Quoc Thong*
DS-04	*WF-02*	*Exception*	*Active replay with controllable packet loss*	*Stop samples; observe timeout; reconnect with two valid samples*	*One communication-loss episode opens and recovery is traceable*	*Le Quoc Thong*
DS-05	*WF-03*	*Main*	*Completed seeded mission with telemetry/incidents*	*Open history, playback, report and analytics*	*Authorized evidence and metrics show correct units/coverage*	*Dang Dao Anh Duc*
DS-06	*WF-03*	*Exception*	*Completed mission with a telemetry gap*	*Generate/open report*	*Missing coverage is explicit and is not reported as zero risk*	*Dang Dao Anh Duc*

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-07	*WF-04*	*Main*	*Authorized mission and configured provider stub*	*Request assessment and inspect provenance*	*Advice is displayed and mission state is unchanged*	*Giang Nguyen Gia Khanh*
DS-08	*WF-04*	*Exception*	*Provider timeout/rate-limit fixture*	*Request assessment; open manual review*	*Unavailable/retry state appears; manual review remains usable*	*Giang Nguyen Gia Khanh*

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	*6*	*3*	*18*
Main workflows (end-to-end)	*4*	*1*	*4*
State transitions of main entities	*12*	*1*	*12*
Committed total			*34*
Automated in CI target		*>= 50%*	*18 (52.9%)*

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

Service and API boundary

MissionAutomationService and MissionAutomationController provide GET /api/missions/automation/due and GET /api/missions/{id}/automation, POST presence/confirm/cancel and GET plan below that mission automation route. Owner authorization and uav_operator role apply. MissionAutomationState retains presence/confirmation, confirmed plan hash and connection ID; trusted enrollment and mission leases remain separate. FlightPreparationPrompt polls only when visible/online. Migration: 20261104115300_gateway_enrollment_and_flight_preparation.sql; deploy backend, frontend and gateway package together.

Diagram maintenance

Existing diagrams are unchanged. Update architecture, ERD, mission state and planning/gateway sequences for enrollment and readiness. Confirmation must not transition the mission to InProgress.